5.4.3 Uniform Continuity Theorem *Let I be a closed bounded interval and let $f : I \to \mathbb{R}$ be continuous on I. Then f is uniformly continuous on I.*

Proof. If f is not uniformly continuous on I then, by the preceding result, there exists $\varepsilon_0 > 0$ and two sequences (x_n) and (u_n) in I such that $|x_n - u_n| < 1/n$ and $|f(x_n) - f(u_n)| \geq \varepsilon_0$ for all $n \in \mathbb{N}$. Since I is bounded, the sequence (x_n) is bounded; by the Bolzano-Weierstrass Theorem 3.4.8 there is a subsequence (x_{n_k}) of (x_n) that converges to an element z. Since I is closed, the limit z belongs to I, by Theorem 3.2.6. It is clear that the corresponding subsequence (u_{n_k}) also converges to z, since

$$|u_{n_k} - z| \leq |u_{n_k} - x_{n_k}| + |x_{n_k} - z|.$$

Now if f is continuous at the point z, then both of the sequences $(f(x_{n_k}))$ and $(f(u_{n_k}))$ must converge to $f(z)$. But this is not possible since

$$|f(x_n) - f(u_n)| \geq \varepsilon_0$$

for all $n \in \mathbb{N}$. Thus the hypothesis that f is not uniformly continuous on the closed bounded interval I implies that f is not continuous at some point $z \in I$. Consequently, if f is continuous at every point of I, then f is uniformly continuous on I. Q.E.D.

Lipschitz Functions

If a uniformly continuous function is given on a set that is not a closed bounded interval, then it is sometimes difficult to establish its uniform continuity. However, there is a condition that frequently occurs that is sufficient to guarantee uniform continuity. It is named after Rudolf Lipschitz (1832–1903) who was a student of Dirichlet and who worked extensively in differential equations and Riemannian geometry.

5.4.4 Definition Let $A \subseteq \mathbb{R}$ and let $f : A \to \mathbb{R}$. If there exists a constant $K > 0$ such that

$$|f(x) - f(u)| \leq K|x - u| \tag{4}$$

for all $x, u \in A$, then f is said to be a **Lipschitz function** (or to satisfy a **Lipschitz condition**) on A.

The condition (4) that a function $f : I \to \mathbb{R}$ on an interval I is a Lipschitz function can be interpreted geometrically as follows. If we write the condition as

$$\left|\frac{f(x) - f(u)}{x - u}\right| \leq K, \qquad x, u \in I, x \neq u,$$

then the quantity inside the absolute values is the slope of a line segment joining the points $(x, f(x))$ and $(u, f(u))$. Thus a function f satisfies a Lipschitz condition if and only if the slopes of all line segments joining two points on the graph of $y = f(x)$ over I are bounded by some number K.

5.4.5 Theorem *If $f : A \to \mathbb{R}$ is a Lipschitz function, then f is uniformly continuous on A.*

Proof. If condition (4) is satisfied, then given $\varepsilon > 0$, we can take $\delta := \varepsilon/K$. If $x, u \in A$ satisfy $|x - u| < \delta$, then

$$|f(x) - f(u)| < K \cdot \frac{\varepsilon}{K} = \varepsilon.$$

Therefore f is uniformly continuous on A. Q.E.D.

5.4.6 Examples **(a)** If $f(x) := x^2$ on $A := [0, b]$, where $b > 0$, then

$$|f(x) - f(u)| = |x + u||x - u| \leq 2b|x - u|$$

for all x, u in $[0, b]$. Thus f satisfies (4) with $K := 2b$ on A, and therefore f is uniformly continuous on A. Of course, since f is continuous and A is a closed bounded interval, this can also be deduced from the Uniform Continuity Theorem. (Note that f does *not* satisfy a Lipschitz condition on the interval $[0, \infty)$.)

(b) Not every uniformly continuous function is a Lipschitz function.

Let $g(x) := \sqrt{x}$ for x in the closed bounded interval $I := [0, 2]$. Since g is continuous on I, it follows from the Uniform Continuity Theorem 5.4.3 that g is uniformly continuous on I. However, there is no number $K > 0$ such that $|g(x)| \leq K|x|$ for all $x \in I$. (Why not?) Therefore, g is not a Lipschitz function on I.

(c) The Uniform Continuity Theorem and Theorem 5.4.5 can sometimes be combined to establish the uniform continuity of a function on a set.

We consider $g(x) := \sqrt{x}$ on the set $A := [0, \infty)$. The uniform continuity of g on the interval $I := [0, 2]$ follows from the Uniform Continuity Theorem as noted in (b). If $J := [1, \infty)$, then if both x, u are in J, we have

$$|g(x) - g(u)| = \left|\sqrt{x} - \sqrt{u}\right| = \frac{|x - u|}{\sqrt{x} + \sqrt{u}} \leq \tfrac{1}{2}|x - u|.$$

Thus g is a Lipschitz function on J with constant $K = \frac{1}{2}$, and hence by Theorem 5.4.5, g is uniformly continuous on $[1, \infty)$. Since $A = I \cup J$, it follows [by taking $\delta(\varepsilon) := \inf\{1, \delta_I(\varepsilon), \delta_J(\varepsilon)\}$] that g is uniformly continuous on A. We leave the details to the reader. □

The Continuous Extension Theorem

We have seen examples of functions that are continuous but not uniformly continuous on open intervals; for example, the function $f(x) = 1/x$ on the interval $(0, 1)$. On the other hand, by the Uniform Continuity Theorem, a function that is continuous on a closed bounded interval is always uniformly continuous. So the question arises: Under what conditions is a function uniformly continuous on a bounded *open* interval? The answer reveals the strength of uniform continuity, for it will be shown that a function on (a, b) is uniformly continuous if and only if it can be defined at the endpoints to produce a function that is continuous on the closed interval. We first establish a result that is of interest in itself.

5.4.7 Theorem *If $f : A \to \mathbb{R}$ is uniformly continuous on a subset A of $\mathbb{R}$ and if (x_n) is a Cauchy sequence in A, then $(f(x_n))$ is a Cauchy sequence in $\mathbb{R}$.*

Proof. Let (x_n) be a Cauchy sequence in A, and let $\varepsilon > 0$ be given. First choose $\delta > 0$ such that if x, u in A satisfy $|x - u| < \delta$, then $|f(x) - f(u)| < \varepsilon$. Since (x_n) is a Cauchy sequence, there exists $H(\delta)$ such that $|x_n - x_m| < \delta$ for all $n, m > H(\delta)$. By the choice of δ, this implies that for $n, m > H(\delta)$, we have $|f(x_n) - f(x_m)| < \varepsilon$. Therefore the sequence $(f(x_n))$ is a Cauchy sequence. Q.E.D.

The preceding result gives us an alternative way of seeing that $f(x) := 1/x$ is not uniformly continuous on $(0, 1)$. We note that the sequence given by $x_n := 1/n$ in $(0, 1)$ is a Cauchy sequence, but the image sequence, where $f(x_n) = n$, is not a Cauchy sequence.

5.4.8 Continuous Extension Theorem *A function f is uniformly continuous on the interval (a, b) if and only if it can be defined at the endpoints a and b such that the extended function is continuous on $[a, b]$.*

Proof. ($\Leftarrow$) This direction is trivial.

($\Rightarrow$) Suppose f is uniformly continuous on (a, b). We shall show how to extend f to a; the argument for b is similar. This is done by showing that $\lim_{x \to c} f(x) = L$ exists, and this is accomplished by using the sequential criterion for limits. If (x_n) is a sequence in (a, b) with $\lim(x_n) = a$, then it is a Cauchy sequence, and by the preceding theorem, the sequence $(f(x_n))$ is also a Cauchy sequence, and so is convergent by Theorem 3.5.5. Thus the limit $\lim(f(x_n)) = L$ exists. If (u_n) is any other sequence in (a, b) that converges to a, then $\lim(u_n - x_n) = a - a = 0$, so by the uniform continuity of f we have

$$\begin{aligned} \lim(f(u_n)) &= \lim(f(u_n) - f(x_n)) + \lim(f(x_n)) \\ &= 0 + L = L. \end{aligned}$$

Since we get the same value L for every sequence converging to a, we infer from the sequential criterion for limits that f has limit L at a. If we define $f(a) := L$, then f is continuous at a. The same argument applies to b, so we conclude that f has a continuous extension to the interval $[a, b]$. Q.E.D.

Since the limit of $f(x) := \sin(1/x)$ at 0 does not exist, we infer from the Continuous Extension Theorem that the function is not uniformly continuous on $(0, b]$ for any $b > 0$. On the other hand, since $\lim_{x \to 0} x \sin(1/x) = 0$ exists, the function $g(x) := x \sin(1/x)$ is uniformly continuous on $(0, b]$ for all $b > 0$.

Approximation†

In many applications it is important to be able to approximate continuous functions by functions of an elementary nature. Although there are a variety of definitions that can be used to make the word "approximate" more precise, one of the most natural (as well as one of the most important) is to require that, at every point of the given domain, the approximating function shall not differ from the given function by more than the preassigned error.

5.4.9 Definition A function $s : [a, b] \to \mathbb{R}$ is called a **step function** if $[a, b]$ is the union of a finite number of nonoverlapping intervals $I_1, I_2, \ldots, I_n$ such that s is constant on each interval, that is, $s(x) = c_k$ for all $x \in I_k$, $k = 1, 2, \ldots, n$.

Thus a step function has only a finite number of distinct values.

For example, the function $s : [-2, 4] \to \mathbb{R}$ defined by

$$s(x) := \begin{cases} 0, & -2 \le x < -1, \\ 1, & -1 \le x \le 0, \\ \frac{1}{2}, & 0 < x < \frac{1}{2}, \\ 3, & \frac{1}{2} \le x < 1, \\ -2, & 1 \le x \le 3, \\ 2, & 3 < x \le 4, \end{cases}$$

is a step function. (See Figure 5.4.3.)

†The rest of this section can be omitted on a first reading of this chapter.

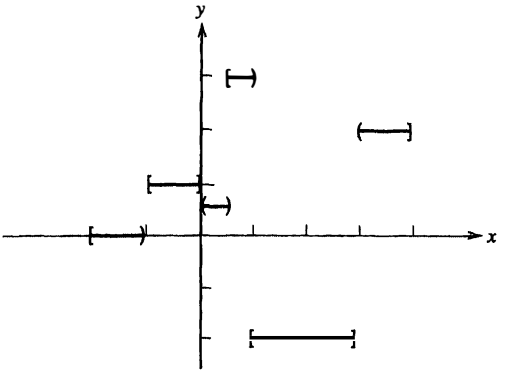


Figure 5.4.3 Graph of $y = s(x)$

We will now show that a continuous function on a closed bounded interval I can be approximated arbitrarily closely by step functions.

5.4.10 Theorem *Let I be a closed bounded interval and let $f : I \to \mathbb{R}$ be continuous on I. If $\varepsilon > 0$, then there exists a step function $s_\varepsilon : I \to \mathbb{R}$ such that $|f(x) - s_\varepsilon(x)| < \varepsilon$ for all $x \in I$.*

Proof. Since (by the Uniform Continuity Theorem 5.4.3) the function f is uniformly continuous, it follows that given $\varepsilon > 0$ there is a number $\delta(\varepsilon) > 0$ such that if $x, y \in I$ and $|x - y| < \delta(\varepsilon)$, then $|f(x) - f(y)| < \varepsilon$. Let $I := [a, b]$ and let $m \in \mathbb{N}$ be sufficiently large so that $h := (b - a)/m < \delta(\varepsilon)$. We now divide $I = [a, b]$ into m disjoint intervals of length h; namely, $I_1 := [a, a + h]$, and $I_k := (a + (k - 1)h, a + kh)$ for $k = 2, \ldots, m$. Since the length of each subinterval I_k is $h < \delta(\varepsilon)$, the difference between any two values of f in I_k is less than ε. We now define

$$s_\varepsilon(x) := f(a + kh) \quad \text{for} \quad x \in I_k, \quad k = 1, \ldots, m, \tag{5}$$

so that s_ε is constant on each interval I_k. (In fact the value of s_ε on I_k is the value of f at the right endpoint of I_k. See Figure 5.4.4.) Consequently if $x \in I_k$, then

$$|f(x) - s_\varepsilon(x)| = |f(x) - f(a + kh)| < \varepsilon.$$

Therefore we have $|f(x) - s_\varepsilon(x)| < \varepsilon$ for all $x \in I$. Q.E.D.

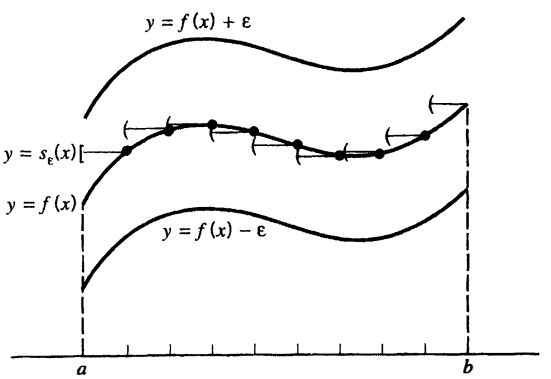


Figure 5.4.4 Approximation by step functions

Note that the proof of the preceding theorem establishes somewhat more than was announced in the statement of the theorem. In fact, we have proved the following, more precise, assertion.

5.4.11 Corollary *Let $I := [a, b]$ be a closed bounded interval and let $f : I \to \mathbb{R}$ be continuous on I. If $\varepsilon > 0$, there exists a natural number m such that if we divide I into m disjoint intervals I_k having length $h := (b - a)/m$, then the step function s_ε defined in equation* (5) *satisfies $|f(x) - s_\varepsilon(x)| < \varepsilon$ for all $x \in I$.*

Step functions are extremely elementary in character, but they are not continuous (except in trivial cases). Since it is often desirable to approximate continuous functions by elementary continuous functions, we now shall show that we can approximate continuous functions by continuous piecewise linear functions.

5.4.12 Definition Let $I := [a, b]$ be an interval. Then a function $g : I \to \mathbb{R}$ is said to be **piecewise linear** on I if I is the union of a finite number of disjoint intervals $I_1, \ldots, I_m$, such that the restriction of g to each interval I_k is a linear function.

Remark It is evident that in order for a piecewise linear function g to be continuous on I, the line segments that form the graph of g must meet at the endpoints of adjacent subintervals I_k, I_{k+1} $(k = 1, \ldots, m - 1)$.

5.4.13 Theorem *Let I be a closed bounded interval and let $f : I \to \mathbb{R}$ be continuous on I. If $\varepsilon > 0$, then there exists a continuous piecewise linear function $g_\varepsilon : I \to \mathbb{R}$ such that $|f(x) - g_\varepsilon(x)| < \varepsilon$ for all $x \in I$.*

Proof. Since f is uniformly continuous on $I := [a, b]$, there is a number $\delta(\varepsilon) > 0$ such that if $x, y \in I$ and $|x - y| < \delta(\varepsilon)$, then $|f(x) - f(y)| < \varepsilon$. Let $m \in \mathbb{N}$ be sufficiently large so that $h := (b - a)/m < \delta(\varepsilon)$. Divide $I = [a, b]$ into m disjoint intervals of length h; namely, let $I_1 = [a, a + h]$, and let $I_k = (a + (k - 1)h, a + kh]$ for $k = 2, \ldots, m$. On each interval I_k we define g_ε to be the linear function joining the points

$$(a + (k - 1)h, f(a + (k - 1)h)) \quad \text{and} \quad (a + kh, f(a + kh)).$$

Then g_ε is a continuous piecewise linear function on I. Since, for $x \in I_k$ the value $f(x)$ is within ε of $f(a + (k - 1)h)$ and $f(a + kh)$, it is an exercise to show that $|f(x) - g_\varepsilon(x)| < \varepsilon$ for all $x \in I_k$; therefore this inequality holds for all $x \in I$. (See Figure 5.4.5.) Q.E.D.

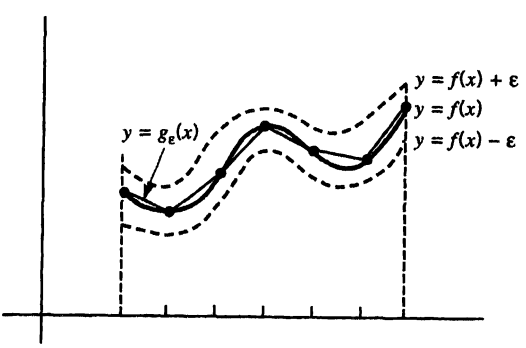


Figure 5.4.5 Approximation by piecewise linear function

We shall close this section by stating the important theorem of Weierstrass concerning the approximation of continuous functions by polynomial functions. As would be expected, in order to obtain an approximation within an arbitrarily preassigned $\varepsilon > 0$, we must be prepared to use polynomials of arbitrarily high degree.

5.4.14 Weierstrass Approximation Theorem *Let $I = [a, b]$ and let $f : I \to \mathbb{R}$ be a continuous function. If $\varepsilon > 0$ is given, then there exists a polynomial function p_ε such that $|f(x) - p_\varepsilon(x)| < \varepsilon$ for all $x \in I$.*

There are a number of proofs of this result. Unfortunately, all of them are rather intricate, or employ results that are not yet at our disposal. (A proof can be found in Bartle, ERA, pp. 169–172, which is listed in the References.)

Exercises for Section 5.4

1. Show that the function $f(x) := 1/x$ is uniformly continuous on the set $A := [a, \infty)$, where a is a positive constant.
2. Show that the function $f(x) := 1/x^2$ is uniformly continuous on $A := [1, \infty)$, but that it is not uniformly continuous on $B := (0, \infty)$.
3. Use the Nonuniform Continuity Criterion 5.4.2 to show that the following functions are not uniformly continuous on the given sets.
   (a) $f(x) := x^2$, $A := [0, \infty)$.
   (b) $g(x) := \sin(1/x)$, $B := (0, \infty)$.
4. Show that the function $f(x) := 1/(1 + x^2)$ for $x \in \mathbb{R}$ is uniformly continuous on $\mathbb{R}$.
5. Show that if f and g are uniformly continuous on a subset A of $\mathbb{R}$, then $f + g$ is uniformly continuous on A.
6. Show that if f and g are uniformly continuous on $A \subseteq \mathbb{R}$ and if they are *both* bounded on A, then their product fg is uniformly continuous on A.
7. If $f(x) := x$ and $g(x) := \sin x$, show that both f and g are uniformly continuous on $\mathbb{R}$, but that their product fg is not uniformly continuous on $\mathbb{R}$.
8. Prove that if f and g are each uniformly continuous on $\mathbb{R}$, then the composite function $f \circ g$ is uniformly continuous on $\mathbb{R}$.
9. If f is uniformly continuous on $A \subseteq \mathbb{R}$, and $|f(x)| \geq k > 0$ for all $x \in A$, show that $1/f$ is uniformly continuous on A.
10. Prove that if f is uniformly continuous on a bounded subset A of $\mathbb{R}$, then f is bounded on A.
11. If $g(x) := \sqrt{x}$ for $x \in [0, 1]$, show that there does not exist a constant K such that $|g(x)| \leq K|x|$ for all $x \in [0, 1]$. Conclude that the uniformly continuous g is not a Lipschitz function on $[0, 1]$.
12. Show that if f is continuous on $[0, \infty)$ and uniformly continuous on $[a, \infty)$ for some positive constant a, then f is uniformly continuous on $[0, \infty)$.
13. Let $A \subseteq \mathbb{R}$ and suppose that $f : A \to \mathbb{R}$ has the following property: for each $\varepsilon > 0$ there exists a function $g_\varepsilon : A \to \mathbb{R}$ such that g_ε is uniformly continuous on A and $|f(x) - g_\varepsilon(x)| < \varepsilon$ for all $x \in A$. Prove that f is uniformly continuous on A.
14. A function $f : \mathbb{R} \to \mathbb{R}$ is said to be **periodic** on $\mathbb{R}$ if there exists a number $p > 0$ such that $f(x + p) = f(x)$ for all $x \in \mathbb{R}$. Prove that a continuous periodic function on $\mathbb{R}$ is bounded and uniformly continuous on $\mathbb{R}$.

15. Let f and g be Lipschitz functions on A.
    (a) Show that the sum $f + g$ is also a Lipschitz function on A.
    (b) Show that if f and g are bounded on A, then the product fg is a Lipschitz function on A.
    (c) Give an example of a Lipschitz function f on $[0, \infty)$ such that its square f^2 is *not* a Lipschitz function.

16. A function is called *absolutely continuous* on an interval I if for any $\varepsilon > 0$ there exists a $\delta > 0$ such that for any pair-wise disjoint subintervals $[x_k, y_k]$, $k = 1, 2, \ldots, n$, of I such that $\sum |x_k - y_k| < \delta$ we have $\sum |f(x_k) - f(y_k)| < \varepsilon$. Show that if f satisfies a Lipschitz condition on I, then f is absolutely continuous on I.

Section 5.5 Continuity and Gauges[†]

We will now introduce some concepts that will be used later—especially in Chapters 7 and 10 on integration theory. However, we wish to introduce the notion of a "gauge" now because of its connection with the study of continuous functions. We first define the notion of a tagged partition of an interval.

5.5.1 Definition A **partition** of an interval $I := [a, b]$ is a collection $\mathcal{P} = \{I_1, \ldots, I_n\}$ of non-overlapping closed intervals whose union is $[a, b]$. We ordinarily denote the intervals by $I_i := [x_{i-1}, x_i]$, where

$$a = x_0 < \cdots < x_{i-1} < x_i < \cdots < x_n = b.$$

The points x_i $(i = 0, \ldots, n)$ are called the **partition points** of $\mathcal{P}$. If a point t_i has been chosen from each interval I_i, for $i = 1, \ldots, n$, then the points t_i are called the **tags** and the set of ordered pairs

$$\dot{\mathcal{P}} = \{(I_1, t_1), \ldots, (I_n, t_n)\}$$

is called a **tagged partition** of I. (The dot signifies that the partition is tagged.)

The "fineness" of a partition $\mathcal{P}$ refers to the lengths of the subintervals in $\mathcal{P}$. Instead of requiring that all subintervals have length less than some specific quantity, it is often useful to allow varying degrees of fineness for different subintervals I_i in $\mathcal{P}$. This is accomplished by the use of a "gauge," which we now define.

5.5.2 Definition A **gauge** on I is a strictly positive function defined on I. If δ is a gauge on I, then a (tagged) partition $\dot{\mathcal{P}}$ is said to be δ-**fine** if

$$(1) \qquad t_i \in I_i \subseteq [t_i - \delta(t_i), t_i + \delta(t_i)] \quad \text{for} \quad i = 1, \ldots, n.$$

We note that the notion of δ-fineness requires that the partition be tagged, so we do not need to say "tagged partition" in this case.

A gauge δ on an interval I assigns an interval $[t - \delta(t), t + \delta(t)]$ to each point $t \in I$. The δ-fineness of a partition $\dot{\mathcal{P}}$ requires that each subinterval I_i of $\dot{\mathcal{P}}$ is contained in the interval determined by the gauge δ and the tag t_i for that subinterval. This is indicated by the inclusions in (1); see Figure 5.5.1. Note that the length of the subintervals is also controlled by the gauge and the tags; the next lemma reflects that control.

[†]This section can be omitted on a first reading of this chapter.



Figure 5.5.1 Inclusion (1)

5.5.3 Lemma *If a partition $\dot{\mathcal{P}}$ of $I := [a, b]$ is δ-fine and $x \in I$, then there exists a tag t_i in $\dot{\mathcal{P}}$ such that $|x - t_i| \leq \delta(t_i)$.*

Proof. If $x \in I$, there exists a subinterval $[x_{i-1}, x_i]$ from $\dot{\mathcal{P}}$ that contains x. Since $\dot{\mathcal{P}}$ is δ-fine, then

$$t_i - \delta(t_i) \leq x_{i-1} \leq x \leq x_i \leq t_i + \delta(t_i), \tag{2}$$

whence it follows that $|x - t_i| \leq \delta(t_i)$. Q.E.D.

In the theory of Riemann integration, we will use gauges δ that are constant functions to control the fineness of the partition; in the theory of the *generalized* Riemann integral, the use of nonconstant gauges is essential. But nonconstant gauge functions arise quite naturally in connection with continuous functions. For, let $f : I \to \mathbb{R}$ be continuous on I and let $\varepsilon > 0$ be given. Then, for each point $t \in I$ there exists $\delta_\varepsilon(t) > 0$ such that if $|x - t| < \delta_\varepsilon(t)$ and $x \in I$, then $|f(x) - f(t)| < \varepsilon$. Since δ_ε is defined and is strictly positive on I, the function δ_ε is a gauge on I. Later in this section, we will use the relations between gauges and continuity to give alternative proofs of the fundamental properties of continuous functions discussed in Sections 5.3 and 5.4.

5.5.4 Examples **(a)** If δ and γ are gauges on $I := [a, b]$ and if $0 < \delta(x) \leq \gamma(x)$ for all $x \in I$, then every partition $\dot{\mathcal{P}}$ that is δ-fine is also γ-fine. This follows immediately from the inequalities

$$t_i - \gamma(t_i) \leq t_i - \delta(t_i) \quad \text{and} \quad t_i + \delta(t_i) \leq t_i + \gamma(t_i)$$

which imply that

$$t_i \in [t_i - \delta(t_i), t_i + \delta(t_i)] \subseteq [t_i - \gamma(t_i), t_i + \gamma(t_i)] \quad \text{for} \quad i = 1, \ldots, n.$$

(b) If δ_1 and δ_2 are gauges on $I := [a, b]$ and if

$$\delta(x) := \min\{\delta_1(x), \delta_2(x)\} \quad \text{for all} \quad x \in I,$$

then δ is also a gauge on I. Moreover, since $\delta(x) \leq \delta_1(x)$, then every δ-fine partition is δ_1-fine. Similarly, every δ-fine partition is also δ_2-fine.

(c) Suppose that δ is defined on $I := [0, 1]$ by

$$\delta(x) := \begin{cases} \frac{1}{10} & \text{if} \quad x = 0, \\ \frac{1}{2}x & \text{if} \quad 0 < x \leq 1. \end{cases}$$

Then δ is a gauge on $[0, 1]$. If $0 < t \leq 1$, then $[t - \delta(t), t + \delta(t)] = \left[\frac{1}{2}t, \frac{3}{2}t\right]$, which does not contain the point 0. Thus, if $\dot{\mathcal{P}}$ is a δ-fine partition of I, then the only subinterval in $\dot{\mathcal{P}}$ that contains 0 must have the point 0 as its tag.

(d) Let γ be defined on $I := [0, 1]$ by

$$\gamma(x) := \begin{cases} \frac{1}{10} & \text{if} \quad x = 0 \text{ or } x = 1, \\ \frac{1}{2}x & \text{if} \quad 0 < x \leq \frac{1}{2}, \\ \frac{1}{2}(1-x) & \text{if} \quad \frac{1}{2} < x < 1. \end{cases}$$

Then γ is a gauge on I, and it is an exercise to show that the subintervals in any γ-fine partition that contain the points 0 or 1 must have these points as tags. □

Existence of δ-Fine Partitions

In view of the above examples, it is not obvious that an arbitrary gauge δ admits a δ-fine partition. We now use the Supremum Property of $\mathbb{R}$ to establish the existence of δ-fine partitions. In the exercises, we will sketch a proof based on the Nested Intervals Theorem 2.5.2.

5.5.5 Theorem *If δ is a gauge defined on the interval $[a, b]$, then there exists a δ-fine partition of $[a, b]$.*

Proof. Let E denote the set of all points $x \in [a, b]$ such that there exists a δ-fine partition of the subinterval $[a, x]$. The set E is not empty, since the pair $([a, x], a)$ is a δ-fine partition of the interval $[a, x]$ when $x \in [a, a + \delta(a)]$ and $x \leq b$. Since $E \subseteq [a, b]$, the set E is also bounded. Let $u := \sup E$ so that $a < u \leq b$. We will show that $u \in E$ and that $u = b$.

We claim that $u \in E$. Since $u - \delta(u) < u = \sup E$, there exists $v \in E$ such that $u - \delta(u) < v < u$. Let $\dot{\mathcal{P}}_1$ be a δ-fine partition of $[a, v]$ and let $\dot{\mathcal{P}}_2 := \dot{\mathcal{P}}_1 \cup ([v, u], u)$. Then $\dot{\mathcal{P}}_2$ is a δ-fine partition of $[a, u]$, so that $u \in E$.

If $u < b$, let $w \in [a, b]$ be such that $u < w < u + \delta(u)$. If $\dot{\mathcal{Q}}_1$ is a δ-fine partition of $[a, u]$, we let $\dot{\mathcal{Q}}_2 := \dot{\mathcal{Q}}_1 \cup ([u, w], u)$. Then $\dot{\mathcal{Q}}_2$ is a δ-fine partition of $[a, w]$, whence $w \in E$. But this contradicts the supposition that u is an upper bound of E. Therefore $u = b$. Q.E.D.

Some Applications

Following R. A. Gordon (see his *Monthly* article in the References), we will now show that some of the major theorems in the two preceding sections can be proved by using gauges.

Alternate Proof of Theorem 5.3.2: Boundedness Theorem. Since f is continuous on I, then for each $t \in I$ there exists $\delta(t) > 0$ such that if $x \in I$ and $|x - t| \leq \delta(t)$, then $|f(x) - f(t)| \leq 1$. Thus δ is a gauge on I. Let $\{(I_i, t_i)\}_{i=1}^n$ be a δ-fine partition of I and let $K := \max\{|f(t_i)| : i = 1, \ldots, n\}$. By Lemma 5.5.3, given any $x \in I$ there exists i with $|x - t_i| \leq \delta(t_i)$, whence

$$|f(x)| \leq |f(x) - f(t_i)| + |f(t_i)| \leq 1 + K.$$

Since $x \in I$ is arbitrary, then f is bounded by $1 + K$ on I. Q.E.D.

Alternate Proof of Theorem 5.3.4: Maximum-Minimum Theorem. We will prove the existence of x^*. Let $M := \sup\{f(x) : x \in I\}$ and suppose that $f(x) < M$ for all $x \in I$. Since f is continuous on I, for each $t \in I$ there exists $\delta(t) > 0$ such that if $x \in I$ and $|x - t| \leq \delta(t)$, then $f(x) < \frac{1}{2}(M + f(t))$. Thus δ is a gauge on I, and if $\{(I_i, t_i)\}_{i=1}^n$ is a δ-fine partition of I, we let

$$\tilde{M} := \tfrac{1}{2}\max\{M + f(t_1), \ldots, M + f(t_n)\}.$$

By Lemma 5.5.3, given any $x \in I$, there exists i with $|x - t_i| \leq \delta(t_i)$, whence

$$f(x) < \tfrac{1}{2}(M + f(t_i)) \leq \tilde{M}.$$

Since $x \in I$ is arbitrary, then $\tilde{M} (< M)$ is an upper bound for f on I, contrary to the definition of M as the supremum of f. Q.E.D.

Alternate Proof of Theorem 5.3.5: Location of Roots Theorem. We assume that $f(t) \neq 0$ for all $t \in I$. Since f is continuous at t, Exercise 5.1.7 implies that there exists $\delta(t) > 0$ such that if $x \in I$ and $|x - t| \leq \delta(t)$, then $f(x) < 0$ if $f(t) < 0$, and $f(x) > 0$ if $f(t) > 0$. Then δ is a gauge on I and we let $\{(I_i, t_i)\}_{i=1}^n$ be a δ-fine partition. Note that for each i, either $f(x) < 0$ for all $x \in [x_{i-1}, x_i]$ or $f(x) > 0$ for all such x. Since $f(x_0) = f(a) < 0$, this implies that $f(x_1) < 0$, which in turn implies that $f(x_2) < 0$. Continuing in this way, we have $f(b) = f(x_n) < 0$, contrary to the hypothesis that $f(b) > 0$. Q.E.D.

Alternate Proof of Theorem 5.4.3: Uniform Continuity Theorem. Let $\varepsilon > 0$ be given. Since f is continuous at $t \in I$, there exists $\delta(t) > 0$ such that if $x \in I$ and $|x - t| \leq 2\delta(t)$, then $|f(x) - f(t)| \leq \frac{1}{2}\varepsilon$. Thus δ is a gauge on I. If $\{(I_i, t_i)\}_{i=1}^n$ is a δ-fine partition of I, let $\delta_\varepsilon := \min\{\delta(t_1), \ldots, \delta(t_n)\}$. Now suppose that $x, u \in I$ and $|x - u| \leq \delta_\varepsilon$, and choose i with $|x - t_i| \leq \delta(t_i)$. Since

$$|u - t_i| \leq |u - x| + |x - t_i| \leq \delta_\varepsilon + \delta(t_i) \leq 2\delta(t_i),$$

then it follows that

$$|f(x) - f(u)| \leq |f(x) - f(t_i)| + |f(t_i) - f(u)| \leq \tfrac{1}{2}\varepsilon + \tfrac{1}{2}\varepsilon = \varepsilon.$$

Therefore, f is uniformly continuous on I. Q.E.D.

Exercises for Section 5.5

1. Let δ be the gauge on $[0, 1]$ defined by $\delta(0) := \frac{1}{4}$ and $\delta(t) := \frac{1}{2}t$ for $t \in (0, 1]$.
   (a) Show that $\dot{\mathcal{P}}_1 = \{([0, \frac{1}{4}], 0), ([\frac{1}{4}, \frac{1}{2}], \frac{1}{2}), ([\frac{1}{2}, 1], \frac{3}{4})\}$ is δ-fine.
   (b) Show that $\dot{\mathcal{P}}_2 = \{([0, \frac{1}{4}], 0), ([\frac{1}{4}, \frac{1}{2}], \frac{1}{2}), ([\frac{1}{2}, 1], \frac{3}{5})\}$ is not δ-fine.
2. Suppose that δ_1 is the gauge defined by $\delta_1(0) := \frac{1}{4}, \delta_1(t) := \frac{3}{4}t$ for $t \in (0, 1]$. Are the partitions given in Exercise 1 δ_1-fine? Note that $\delta(t) \leq \delta_1(t)$ for all $t \in [0, 1]$.
3. Suppose that δ_2 is the gauge defined by $\delta_2(0) := \frac{1}{10}$ and $\delta_2(t) := \frac{9}{10}t$ for $t \in (0, 1]$. Are the partitions given in Exercise 1 δ_2-fine?
4. Let γ be the gauge in Example 5.5.4(d).
   (a) If $t \in (0, \frac{1}{2}]$ show that $[t - \gamma(t), t + \gamma(t)] = [\frac{1}{2}t, \frac{3}{2}t] \subseteq (0, \frac{3}{4}]$.
   (b) If $t \in (\frac{1}{2}, 1)$ show that $[t - \gamma(t), t + \gamma(t)] \subseteq (\frac{1}{4}, 1)$.
5. Let $a < c < b$ and let δ be a gauge on $[a, b]$. If $\dot{\mathcal{P}}'$ is a δ-fine partition of $[a, c]$ and if $\dot{\mathcal{P}}''$ is a δ-fine partition of $[c, b]$, show that $\dot{\mathcal{P}}' \cup \dot{\mathcal{P}}''$ is a δ-fine partition of $[a, b]$ having c as a partition point.